

Set Algebra on $\mathbb{R}$ Using Intervals

A Problem-Based Study Module (First-Year College)

Learning Goals

By the end of this module, you should be able to:

- Compute unions, intersections, differences, and complements of intervals on $\mathbb{R}$.
- Use De Morgan's laws to simplify complements of complicated set expressions.
- Present final answers as clean unions of intervals (and check endpoints correctly).
- Visualize sets and their complements on number-line diagrams.

1. Notation and the “Universe” Idea

A **universe** U is the set in which all complements are taken.

- Most interval problems use $U = \mathbb{R}$. Then $A^c = \mathbb{R} \setminus A$.
- Sometimes the universe is restricted, e.g. $U = [0, 10]$. Then $A^c = [0, 10] \setminus A$.

Always identify U first; it changes the complement.

2. Toolkit: The Identities You Use Repeatedly

2.1 Membership Dictionary (Turn Sets into Logic)

For any real number x :

$$x \in A \cup B \iff (x \in A) \vee (x \in B), \quad x \in A \cap B \iff (x \in A) \wedge (x \in B).$$

$$x \in A \setminus B \iff (x \in A) \wedge (x \notin B), \quad x \in A^c \iff x \notin A.$$

This is the safest method to avoid endpoint mistakes.

2.2 De Morgan's laws (The Main Engine)

$$(A \cup B)^c = A^c \cap B^c, \quad (A \cap B)^c = A^c \cup B^c.$$

(Also valid for more than two sets.)

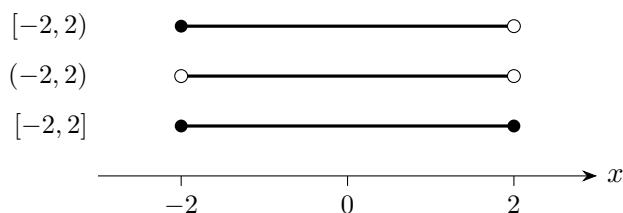
2.3 Complements of Basic Intervals in $\mathbb{R}$

Let $a < b$. Then:

$$(a, b)^c = (-\infty, a] \cup [b, \infty), \quad [a, b]^c = (-\infty, a) \cup (b, \infty),$$

$$[a, b)^c = (-\infty, a) \cup [b, \infty), \quad (a, b]^c = (-\infty, a] \cup (b, \infty).$$

Open/closed endpoint meaning (picture)



Closed dot = included endpoint; open circle = excluded endpoint.

3. Worked Problems (Diagrams and Detailed Steps)

Problem

Let $A = [-1, 3)$ and $B = (2, 5]$ in $U = \mathbb{R}$. Find $(A \cup B)^c$.

Solution (step-by-step)

Step 1: Understand A and B .

$$A = [-1, 3) \text{ means } -1 \leq x < 3, \quad B = (2, 5] \text{ means } 2 < x \leq 5.$$

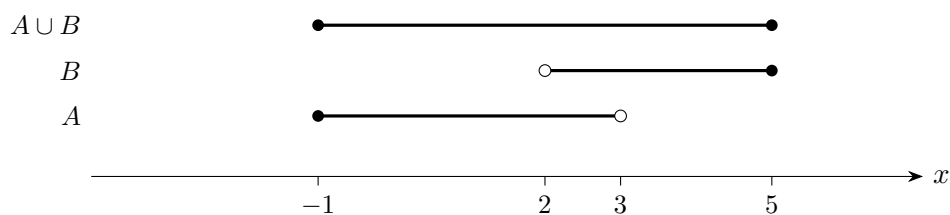
Step 2: Compute the Union $A \cup B$.

Because A reaches up to (but not including) 3 and B starts just above 2, the sets *overlap* on $(2, 3)$. Together they cover continuously from -1 up to 5 (including 5). Also, 3 is included because $3 \in B$. Hence

$$A \cup B = [-1, 5].$$

Step 3: Take Complement in $\mathbb{R}$.

$$(A \cup B)^c = \mathbb{R} \setminus [-1, 5] = (-\infty, -1) \cup (5, \infty).$$



Problem

Let $A = (0, 2]$ and $B = [4, 6)$ in $U = \mathbb{R}$. Find $(A \cup B)^c$.

Solution (step-by-step)

Step 1: Compute the union. Here A and B are disjoint (there is a gap between 2 and 4), so

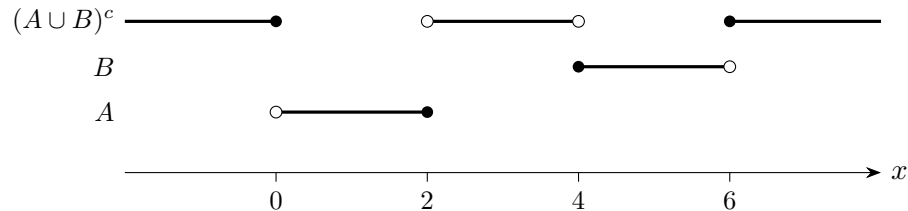
$$A \cup B = (0, 2] \cup [4, 6).$$

Step 2: Complement in $\mathbb{R}$ = everything not in that union. There are three regions:

- left of 0 (including 0 because $0 \notin (0, 2]$),
- between 2 and 4 (excluding 2 because $2 \in A$, including 4 because $4 \in B$),
- right of 6 (including 6 because $6 \notin [4, 6)$).

Thus

$$(A \cup B)^c = (-\infty, 0] \cup (2, 4) \cup [6, \infty).$$



Problem

Let $A = (-3, 1]$ and $B = [-1, 4)$ in $U = \mathbb{R}$. Find $(A \cap B)^c$.

Solution (step-by-step)

Step 1: Compute the Intersection $A \cap B$.

Intersection keeps points that lie in both sets.

$$A = (-3, 1], \quad B = [-1, 4).$$

So:

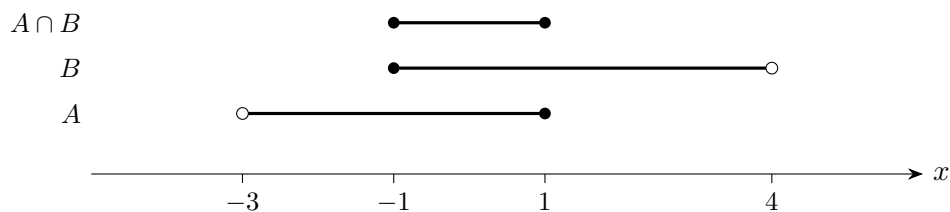
- the left endpoint becomes -1 (since B starts at -1 and A already contains -1),
- the right endpoint becomes 1 (since A ends at 1 and B contains 1).

Hence

$$A \cap B = [-1, 1].$$

Step 2: Complement in $\mathbb{R}$.

$$(A \cap B)^c = \mathbb{R} \setminus [-1, 1] = (-\infty, -1) \cup (1, \infty).$$



Problem

Let $U = [0, 10]$, $A = [2, 7]$, $B = (5, 9]$. Find $(A \cup B)^c$ relative to U .

Solution (step-by-step)

Step 1: Compute $A \cup B$ Inside $\mathbb{R}$.

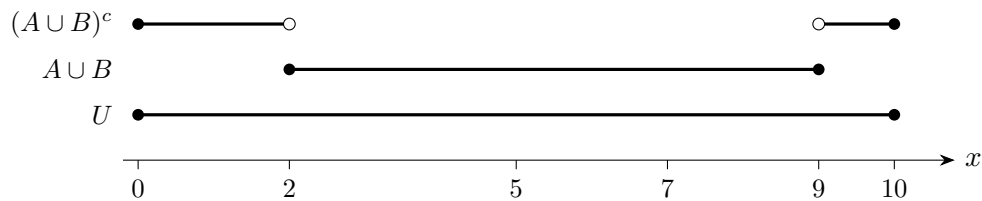
A covers from 2 up to 7 (excluding 7). B covers from 5 (excluding 5) up to 9 (including 9). So together they cover $[2, 9]$ (note: 7 is included because $7 \in B$).

$$A \cup B = [2, 9].$$

Step 2: Complement *relative to* $U = [0, 10]$.

$$(A \cup B)^c = U \setminus (A \cup B) = [0, 10] \setminus [2, 9] = [0, 2) \cup (9, 10].$$

Key point: We are not allowed to go outside $[0, 10]$ when taking the complement.



Problem

Let $A = (-\infty, 3)$ and $B = (1, \infty)$ in $U = \mathbb{R}$. Find $A \setminus B$, $B \setminus A$, and $A \triangle B$.

Solution (step-by-step)

Step 1: Compute $A \setminus B$.

$$A \setminus B = \{x : x < 3 \text{ and } x \notin (1, \infty)\}.$$

But $x \notin (1, \infty)$ means $x \leq 1$. So

$$A \setminus B = (-\infty, 1].$$

Step 2: Compute $B \setminus A$.

$$B \setminus A = \{x : x > 1 \text{ and } x \notin (-\infty, 3)\}.$$

Now $x \notin (-\infty, 3)$ means $x \geq 3$. So

$$B \setminus A = [3, \infty).$$

Step 3: Symmetric difference is “exactly one.”

$$A \triangle B = (A \setminus B) \cup (B \setminus A) = (-\infty, 1] \cup [3, \infty).$$

Problem

Let $A = [-2, 4]$ and $B = (0, 3)$ in $U = \mathbb{R}$. Find $(A \setminus B)^c$.

Solution (step-by-step)

Step 1: Compute $A \setminus B$ by removing $(0, 3)$ from $[-2, 4]$.

Inside $[-2, 4]$, the open interval $(0, 3)$ removes all points strictly between 0 and 3. The endpoints 0 and 3 remain because $0, 3 \notin (0, 3)$. So

$$A \setminus B = [-2, 0] \cup [3, 4].$$

Step 2: Complement in $\mathbb{R}$. The complement of a union is the intersection of complements (De Morgan), but here it is easiest to reason directly: everything outside $[-2, 0] \cup [3, 4]$ equals

$$(A \setminus B)^c = (-\infty, -2) \cup (0, 3) \cup (4, \infty).$$

Checkpoint: 0 is *in* $A \setminus B$, so 0 is *not* in the complement; hence $(0, 3)$ is correct (open at 0).

Problem

Let $A = [-5, -1) \cup (2, 4]$ in $U = \mathbb{R}$. Find A^c .

Solution (step-by-step)

Step 1: Identify the “gaps” in $\mathbb{R}$.

The set A occupies two blocks:

$$[-5, -1) \quad \text{and} \quad (2, 4].$$

So the complement is everything:

- left of -5 (excluding -5 because $-5 \in A$),
- between -1 and 2 (including both endpoints because $-1 \notin A$ and $2 \notin A$),
- right of 4 (excluding 4 because $4 \in A$).

Therefore

$$A^c = (-\infty, -5) \cup [-1, 2] \cup (4, \infty).$$

Problem

Let $A = (-2, 5)$ and $B = [1, \infty)$ in $U = \mathbb{R}$. Find $(A \cap B)^c$ and $(A \cup B)^c$.

Solution (step-by-step)

Part (i): $(A \cap B)^c$.

First compute the intersection:

$$A \cap B = (-2, 5) \cap [1, \infty) = [1, 5).$$

Now complement in $\mathbb{R}$:

$$(A \cap B)^c = \mathbb{R} \setminus [1, 5) = (-\infty, 1) \cup [5, \infty).$$

Part (ii): $(A \cup B)^c$.

Union:

$$A \cup B = (-2, 5) \cup [1, \infty) = (-2, \infty).$$

Complement:

$$(A \cup B)^c = \mathbb{R} \setminus (-2, \infty) = (-\infty, -2].$$

Problem

Let $A = [-1, 2)$ and $B = (3, 5]$ in $U = \mathbb{R}$. Describe $S = (A \cup B)^c$ as a union of intervals.

Solution (step-by-step)

Step 1: Union.

$$A \cup B = [-1, 2) \cup (3, 5].$$

Step 2: Complement in $\mathbb{R}$.

- Left of -1 (excluding -1 since $-1 \in A$): $(-\infty, -1)$.
- Between 2 and 3 : here $2 \notin A$ and $3 \notin B$, so both endpoints are included: $[2, 3]$.
- Right of 5 (excluding 5 since $5 \in B$): $(5, \infty)$.

Thus

$$S = (-\infty, -1) \cup [2, 3] \cup (5, \infty).$$

Problem

Let $A = [0, 1)$ and $B = (2, 3]$ in $U = \mathbb{R}$. Compute $(A^c \cap B^c)^c$.

Solution (step-by-step)

Step 1: Use De Morgan (work symbolically first).

$$(A^c \cap B^c)^c = (A^c)^c \cup (B^c)^c = A \cup B.$$

This step is the main idea: the outer complement flips $\cap$ to $\cup$ and removes complements.

Step 2: Compute $A \cup B$.

$$A \cup B = [0, 1) \cup (2, 3].$$

So

$$(A^c \cap B^c)^c = [0, 1) \cup (2, 3].$$

4. Common Mistakes (How to Self-Check)

- **Forgetting the universe:** A^c means $U \setminus A$, not always $\mathbb{R} \setminus A$.
- **Endpoint errors:** check each special point (a, b) by testing membership.
- **Complement of a union:** do *not* complement each interval piece without using De Morgan correctly.
- **Self-check trick:** pick one point from each region of your final answer and verify it should be in/out.

5. Practice Exercises

Instructions

For each exercise, simplify the expression and write your final answer as a **union of disjoint intervals**. Always state the universe U used for complements.

1. ($U = \mathbb{R}$) $A = [-4, 1)$, $B = (0, 6]$. Find $(A \cup B)^c$.
2. ($U = \mathbb{R}$) $A = (-\infty, -2]$, $B = [-1, 3)$. Find $(A \cap B)^c$.
3. ($U = \mathbb{R}$) $A = (1, 5]$, $B = [3, 7)$. Find $A \setminus B$.
4. ($U = \mathbb{R}$) $A = [-3, 2]$, $B = (-1, 4)$. Find $B \setminus A$.
5. ($U = \mathbb{R}$) $A = (-2, 3)$, $B = (3, 8]$. Find $A \triangle B$.
6. ($U = [-5, 5]$) $A = [-5, -1) \cup (1, 5]$. Find A^c in U .
7. ($U = [0, 12]$) $A = (2, 9)$, $B = [6, 12]$. Find $(A \cup B)^c$ in U .
8. ($U = \mathbb{R}$) $A = [0, 4)$, $B = (2, 6]$, $C = [-1, 1]$. Find $(A \cap B) \cup C$.
9. ($U = \mathbb{R}$) $A = [-2, 5]$, $B = (0, 3]$, $C = [3, 7)$. Find $A \setminus (B \cup C)$.
10. ($U = \mathbb{R}$) $A = (-\infty, 2)$, $B = [1, 4)$, $C = (3, \infty)$. Find $(A \cup C)^c \cap B$.
11. ($U = \mathbb{R}$) Prove using interval reasoning that $(A \setminus B)^c = A^c \cup B$ for $A = [-1, 3]$, $B = (0, 2)$ by computing both sides explicitly.
12. ($U = \mathbb{R}$) $A = [-5, -2) \cup [0, 1)$, find A^c .

6. Answer key

Answer key (final forms only)

(Shown for quick checking; try the exercises first.)

1. $(-\infty, -4) \cup [1, \infty)$
2. $(-\infty, -1) \cup [3, \infty)$
3. $(1, 3)$
4. $(2, 4)$
5. $(-2, 3) \cup (3, 8]$
6. $[-1, 1]$
7. $[0, 2] \cup [12, 12]$ (i.e. $[0, 2] \cup \{12\}$)
8. $[-1, 1] \cup (2, 4)$
9. $[-2, 0] \cup \{5\}$ (i.e. $[-2, 0] \cup [5, 5]$)
10. $[1, 2)$
11. (Compute to verify equality; both sides simplify to $(-\infty, -1) \cup (0, 2) \cup [3, \infty)$)
12. $(-\infty, -5) \cup [-2, 0) \cup [1, \infty)$

Suggestion for exam practice: For any final answer, test the endpoints and one interior point from each interval piece.